

[SINGLE CORRECT CHOICE TYPE]

- Q.1 Two heavy bodies of mass 'm' and '3m' tied together with a light string of mass density μ are dropped from a helicopter as shown. A constant air friction force 'F' acts on both of them. If the length of the string is ℓ . Determine the time taken by a pulse to go from top to bottom of the string.



- (A) $\ell \sqrt{\frac{\mu}{F}}$ (B) $\ell \sqrt{\frac{2\mu}{F}}$ (C) $\ell \sqrt{\frac{4\mu}{F}}$ (D) $\ell \sqrt{\frac{\mu}{2F}}$
- Q.2 A transverse wave is described by the equation $Y = A \sin^2(\omega t - kx)$. The maximum particle velocity is equal to the wave velocity, if the value of A is
(A) $3/k$ (B) $2/k$ (C) $1/k$ (D) none of these
- Q.3 Wave in string is represented by : $y(x, t) = 2 \sin^2(k_1 x - \omega_1 t) \cos^2(k_2 x - \omega_2 t)$. The numbers of component waves are
(1) two (2) three (3) four (4) six
- Q.4 A revolving ring in a siren revolves 420 times per minute and it produces 2 beats per second with a tuning fork of frequency 348 Hz. Find the possible number of holes in the ring.
(A) 49 (B) 50 (C) 36 (D) 48
- Q.5 In a resonance tube experiment, the first two resonances are observed at length 10.5 cm and 29.5 cm. The third resonance is observed at the length _____ (cm).
(A) 47.5 (B) 58.5 (C) 48.5 (D) 82.8
- Q.6 A steel wire of length 1 m and mass 1 gm and having a uniform cross-sectional area of 10^{-6} m^2 is rigidly fixed at both the ends. The temperature of the wire is lowered by 20°C . If the transverse standing wave are set up by plucking the string in the middle, the frequency of the fundamental mode of vibration is ($Y_{\text{steel}} = 2 \times 10^{11} \text{ N/m}^2$, $\alpha_{\text{steel}} = 1.21 \times 10^{-5} / ^\circ\text{C}$)
(A) 110 Hz (B) 1.1 Hz (C) 1100 Hz (D) 11 Hz
- Q.7 A snapshot of a travelling wave is shown in figure-1 and a snapshot of a standing wave is shown in figure-2. (Particles are at their respective extremes. The amplitude, frequency, and the speed of both the waves is the same. The string in both the cases is identical). If the energy per unit length of system at point A in figure-1, is x_1 and that in figure-2 is x_2 . What is value of $\frac{x_1}{x_2}$?

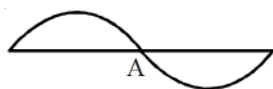


Fig.-1

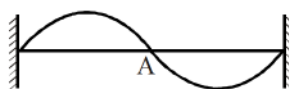
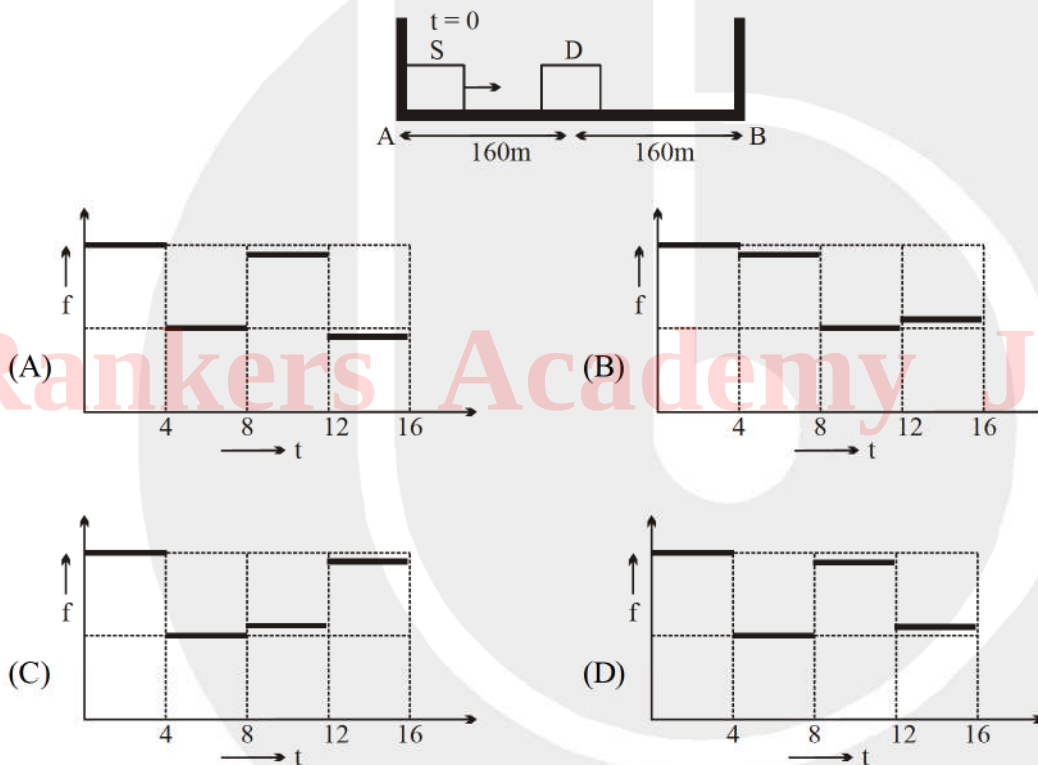


Fig.-2

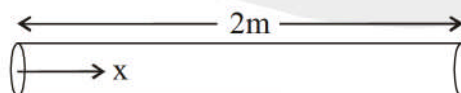
- (A) 2 : 1 (B) 1 : 2 (C) 1 : 1 (D) 3 : 1

- Q.8 A standing wave is produced in a vapour of an element of atomic mass $50 \text{ (g mol}^{-1}\text{)}$ at 400 K . Nodes are found to be $\frac{5}{3} \text{ m}$ apart when frequency of source was 100 Hz . The vapour used for the experiment is ($R = \frac{25}{3} \text{ JK}^{-1} \text{ mol}^{-1}$)
- (A) Monatomic (B) Diatomic (C) Polyatomic (D) Can not be determined
- Q.9 Equation of a longitudinal wave is given as $y = 10^{-2} \sin 2\pi \left(1000t + \frac{50x}{17} \right)$ (all SI units). At $t = 0$, change in pressure is maximum (in modulus) at $x =$
- (A) 0.34 (B) 0.255 (C) 0.085 (D) All of these
- Q.10 A small source S of mass m starts moving with velocity 40 m/s towards a detector D of same mass m on a frictionless surface AB as shown in the figure. The frequency received by the detector as a function of time in the time interval $0 \leq t \leq 16 \text{ s}$ is: [Assume all collisions to be elastic]. Take $v_s = 340 \text{ m/s}$.



[MULTIPLE CORRECT CHOICE TYPE]

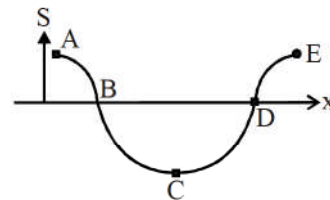
- Q.11 An open organ pipe of radius 1 cm is vibrating in its 1^{st} overtone. Here origin is at left end of the pipe.



The x coordinate of pressure node can be

- (A) $+0.6 \text{ cm}$ (B) -0.6 cm (C) 100.6 cm (D) 100 cm

- Q.12 The figure represents a displacement versus position graph for a longitudinal wave travelling in positive x-direction. Then
 (A) part ABC represents region of compression
 (B) part ABC represents region of rarefaction
 (C) part CDE represents region of compression
 (D) part CDE represents region of rarefaction



- Q.13 Standing sound waves produced in a pipe that is 1.20m long. For the first overtone determine the locations along the pipe (measured from the left end) of the displacement nodes (S_N) and the pressure nodes (P_N) if
 (A) The pipe is open at both ends, S_N at 0.3 m, 0.9 m
 (B) The pipe is open at both ends, P_N at 0 m, 0.6m, 1.2 m
 (C) The pipe is closed at the left end and open at the right end S_N at 0m, 0.8m,
 (D) The pipe is closed at the left end and open at the right end P_N at 0.4 m, 1.2 m.

- Q.14 Two narrow cylindrical pipes A and B are filled with hydrogen and helium gases respectively at same temperature. Pipe A is open at both ends and pipe B is open at one end and closed at the other end. If frequency of the second overtone of pipe A is equal to the frequency of the first overtone of the pipe B,

(A) ratio of the velocities of sound waves in the two pipes $\frac{v_A}{v_B} = \frac{1}{2}$

(B) $\frac{v_A}{v_B} = \sqrt{\frac{42}{25}}$

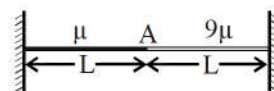
(C) ratio of their lengths $\frac{l_A}{l_B} = \sqrt{\frac{168}{25}}$

(D) $\frac{l_A}{l_B} = 2\sqrt{2}$

- Q.15 A narrow steel rod of length 'L' is rigidly clamped at its mid-point and transverse standing waves of frequency 'f' are set up in it. The speed of transverse waves in the rod is 'c'. Then,
 (A) The free ends of the rod must be antinodes.
 (B) The fundamental frequency of the rod is $c/(L)$
 (C) The second overtone frequency of the rod is $5c/(2L)$
 (D) 'f' can be any integral multiple of the fundamental frequency.

- Q.16 Two sources are placed two wavelengths apart. Point P is a minima in the interference pattern of the two sources. Point Q is a maxima in the interference pattern.
 (A) If one source is turned off, the energy arriving at P increases.
 (B) If one source is turned off, the energy arriving at Q decreases.
 (C) If one source is turned off, the energy arriving at P may decrease.
 (D) If one source is turned off, the energy arriving at Q may increase.

- Q.17 A light string is tied at one end to fixed support and to a heavy string of equal length L at the other end as shown in figure. Mass per unit length of the strings are μ and 9μ and the tension is T. Find the possible values of frequencies such that point A is an antinode.



(A) $\frac{1}{4L} \sqrt{\frac{T}{\mu}}$

(B) $\frac{3}{4L} \sqrt{\frac{T}{\mu}}$

(C) $\frac{3}{2L} \sqrt{\frac{T}{\mu}}$

(D) $\frac{7}{6L} \sqrt{\frac{T}{\mu}}$

[MATRIX TYPE]

- Q.18 Column-I shows different sets of standing waves in a string of length L whose ends are fixed or free according to respective figure & Column-II shows possible equations for them where symbols have usual meaning..

Column-I



Column-II

(P) $y = A \sin \frac{\pi x}{L} \cos \omega t$

(Q) $y = A \sin \frac{2\pi x}{L} \cos \omega t$

(R) $y = A \cos \frac{\pi x}{2L} \cos \omega t$

(S) $y = A \cos \frac{\pi x}{L} \cos \omega t$

- Q.19 Two waves given by equation $y_1 = A \sin (\omega t - kx)$ and $y_2 = 4A \sin (2\omega t - 4kx)$ are traveling in two different strings having same mass per unit length and same cross-sectional area. Tension in first string is four times tension in the second string.

Column-I

- (A) Ratio of energy density U_1/U_2
 (B) Ratio of intensity I_1/I_2
 (C) Ratio of power P_1/P_2
 (D) Ratio of velocity of wave v_1/v_2

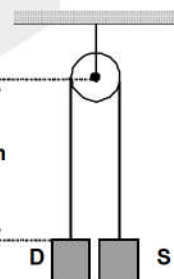
Column-II

- (P) Less than 1
 (Q) $1/32$
 (R) $1/64$
 (S) Greater than 1
 (T) $1/2$

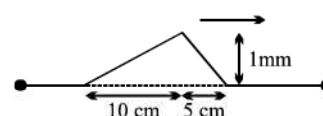
[SUBJECTIVE TYPE]

- Q.20 Sound wave travelling in air ($\gamma = 1.4$, $T = 294$ K, $M = 28$ gm, $R = \frac{25}{3}$ J/mol K) is normally incident on water surface ($B = 2.1025 \times 10^9$ Pa, density $= 10^3$ kg/m³). The pressure amplitude of incident wave is 18 Pa. What is the pressure amplitude (in Pa) of transmitted wave ?

- Q.21 Consider the shown arrangement where a source of sonic oscillation having frequency 500 Hz and a detector are connected to the two free ends of a string. Take the masses of the detector and the source as 4 kg and 6 kg respectively. Initially length of string on both sides of pulley equals 1 m. The system is released from rest. Find the frequency (in Hz) registered by the detector just before it strikes the pulley. The diameter of pulley is small enough to consider the motion of the bodies along the same line. Take velocity of sound $= 318$ m/s. Round off to nearest integer.



- Q.22 Find the kinetic energy of pulse travelling in a taut string.
 Given $T = 10$ N & $\mu = 0.1$ kg/m.



ANSWER KEY

Q.1	B	Q.2	C	Q.3	3	Q.4	A	Q.5	1	Q.6	A	Q.7	A
Q.8	A	Q.9	A	Q.10	D	Q.11	B, D	Q.12	A, D	Q.13	A, B, C, D		
Q.14	B, C	Q.15	A, C	Q.16	B, C	Q.17	A, B	Q.18	(A) – Q ; (B) – P ; (C) – R ; (D) – S				
Q.19	(A) PR (B) PQ (C) PQ (D) S					Q.20.	0029			Q.21	$6f_0/5, 6f_0/7$		
Q.22.	$1.5 \times 10^{-4} \text{ J}$												

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